

Assignment 2

Wireless Protocol Investigation Lab

Evidence-Based 802.11 Security Analysis

Wireless and Mobile Network Security

Department of Computer Science

Course #: 2-7038910-1

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1 Background

Assignment 1 focused on implementation: you built a working Evil Twin attack tool, including wireless discovery, targeted disconnection, rogue access point operation, credential-capture logic, and a defense component. Assignment 2 has a different purpose.

This assignment is not primarily about building another attack tool. It is about becoming a wireless protocol investigator. You will build a controlled Wi-Fi experiment, capture real 802.11 evidence, dissect the captured traffic, and prove a security claim using frames, fields, state transitions, and carefully justified reasoning.

Main Idea

Your task is to investigate a real 802.11 security property using monitor-mode captures, protocol parsing, controlled experiments, and evidence-based analysis. The required output is a technical investigation report supported by PCAP files, annotated packet traces, parser output, and a clear explanation of what the evidence proves.

The final result does **not** have to be a working exploit, attack framework, or polished proof-of-concept. A strong submission is one that can answer questions such as:

- Which exact packet supports your claim?
- Which field are you relying on?
- Is this field protected, authenticated, encrypted, or visible in clear text?
- What changed between your baseline capture and your modified-condition capture?
- What can an attacker infer from this evidence?
- What can an attacker **not** infer from this evidence?

2 Assignment Objectives

By completing this assignment, you will develop the ability to:

- **Analyze 802.11 behavior empirically:** Capture real wireless traffic and connect packet-level observations to protocol behavior.
- **Dissect security-relevant fields:** Identify and interpret 802.11 management frames, RSN information elements, EAPOL messages, WPA2/WPA3/OWE indicators, PMF flags, and PHY/MAC metadata where relevant.
- **Separate observation from interpretation:** Distinguish between what was directly observed in the trace, what can be inferred, and what remains unproven.
- **Understand wireless trust assumptions:** Explain what Wi-Fi clients authenticate, what remains unauthenticated, and what metadata remains visible even when payload encryption is enabled.
- **Produce reproducible evidence:** Submit captures, scripts, and documentation that allow another student or examiner to reproduce your reasoning.

- **Practice graduate-level cybersecurity analysis:** Move beyond tool operation and demonstrate careful reasoning about protocols, threats, and limitations.

3 Important Notes

- **Group work:** The assignment may be performed in groups according to the course instructions.
- **Frontal defense:** The work group meets the instructor in a single Zoom session of approximately 90–120 minutes that covers both Assignment 1 and Assignment 2. Each student is questioned individually within that session. See Section 9 for a detailed description of what the defense involves.
- **No requirement for exploit success:** A working attack or weaponized proof-of-concept is **not** required. The grade is based on the quality of the investigation, evidence, parsing, and reasoning.
- **Your own evidence is mandatory:** Each group must produce its own PCAP/P-CAPNG captures. Captures must not be taken from public datasets, published papers, prior years' submissions, or other groups. PCAP metadata will be verified. For authorization and scope rules, see the warning below.
- **Independent analysis is mandatory:** You may use public sources, documentation, and tools such as Wireshark, TShark, Scapy, and Linux utilities. However, your submitted analysis, packet annotations, parser, and report must be your own work.
- **Hardware variability is expected:** Different adapters, drivers, routers, and clients may expose different behavior. Your limitations section must honestly describe what your hardware could and could not capture.

SEVERE WARNING — Ethical and Legal Scope of the Lab

Authorization. All experiments must be conducted exclusively on networks, devices, channels, and equipment that you own or have explicit permission to test. Unauthorized wireless experimentation is illegal and may result in course disqualification and additional consequences.

Privacy and scope of analysis. Analyze packets only from devices in your own lab setup. Do not target or analyze any data belonging to anyone outside your authorized lab. Frames from other devices that incidentally appear in the capture should be filtered out, and the report must state explicitly that they were not part of your analysis.

4 Assignment Overview

Each group must choose one approved investigation track from Section 5. Multiple groups may choose the same track, but no two groups may submit the same investigative claim, experiment design, and capture scenario. Submit your track selection to the instructor by **[deadline TBA]**. Selections are confirmed by reply email only.

Every group follows the same evidence-based investigation structure:

1. Define a security claim or research question.
2. State the investigative claim you intend to prove before collecting any evidence.
3. Build a controlled Wi-Fi lab.
4. Capture at least two conditions: a baseline condition and a modified condition.

Controlled Variable Requirement

Your baseline and modified-condition captures must differ by one clearly stated experimental variable whenever possible. Examples: PMF disabled vs PMF required, WPA2 vs WPA3-SAE, same SSID vs different SSID, static client vs moving client. Do not change multiple unrelated variables unless you explain why this was unavoidable.

5. Annotate the important packets and fields.
6. Write a small parser or analysis script.
7. Explain the relevant protocol state transitions.
8. Connect every major claim to packet numbers, parsed fields, script output, or cited technical references.

The Organizing Principle

Assignment 2 is not “build another attack.” Assignment 2 is “prove a wireless security claim from packets.”

Evidence Depends on the Claim

The required evidence is determined by the investigative claim. Do not treat the examples in each track as a checklist to reproduce. A valid submission must define the evidence needed for its own claim and then show that the collected traces actually contain that evidence.

Before collecting evidence, each group must prepare a claim-specific evidence plan. The plan must answer:

- What exact claim are you trying to prove, disprove, or qualify?
- Which packets or frame types must appear in the capture for this claim to be testable?
- Which fields must be extracted and interpreted?
- What must change between the baseline condition and the modified condition?
- What parser output, table, or plot will support the claim?
- What result would weaken or contradict the claim?

A strong submission explains not only what was captured, but why those packets and fields are sufficient to support the stated claim.

5 Approved Investigation Tracks

Choose **one** of the following tracks. If you wish to investigate a different topic, you must receive written approval before beginning.

5.1 Track 1: Management-Frame Trust and PMF

Central Question

When does 802.11 trust management frames, and when are those frames protected?

What This Track Investigates

This track continues naturally from Assignment 1. In the Evil Twin assignment, deauthentication and management-frame manipulation were used as practical attack components. In this track, you investigate the protocol reason such behavior is possible and how Protected Management Frames (PMF / 802.11w) changes the picture.

Investigative Claim to Prove

Before collecting evidence, your group must state one precise claim such as: *“We will demonstrate that deauthentication frames are transmitted in clear text and accepted without verification under WPA2 with PMF disabled, and that enabling PMF changes the observable behavior of post-association management frames in a specific, measurable way.”* Your report must return to this claim and prove or qualify it using frame-level evidence.

Required Experiment

Create a controlled lab network and compare at least two of the following conditions:

- PMF disabled.
- PMF capable / optional.
- PMF required, if supported by your AP and client.

Depending on your investigative claim, relevant evidence may include:

- Beacon frames.
- Probe responses.
- Authentication frames.
- Association request and association response.
- RSN information elements.
- Deauthentication and/or disassociation behavior.
- PMF capability and requirement bits.

Required Security Analysis

Your report must answer:

- Which management frames are visible before association?
- Which management frames are unauthenticated?
- Which robust management frames are protected after association and key establishment, and which management frames remain outside PMF protection?
- Where are the PMF Capable and PMF Required bits advertised?
- Why did deauthentication work in Assignment 1?
- Why does PMF not solve every Evil Twin-style problem?
- What remains exposed even when PMF is enabled?

What to Expect

This is a suitable default track for most groups. A WPA2-capable AP is sufficient for the PMF-disabled baseline. To compare PMF capable or PMF required behavior, your AP and client must both support PMF. Existing tools for sending deauthentication frames are common, but your grade is not based on using them. Your grade is based on explaining the captured protocol behavior.

5.2 Track 2: WPA2/WPA3/OWE Handshake Autopsy

Central Question

What exactly happens when a station becomes a secure member of a Wi-Fi network?

What This Track Investigates

This track focuses on authentication, association, RSN negotiation, and key establishment. It prepares you to understand attacks such as KRACK, Dragonblood, PMKID capture, downgrade attacks, and OWE identity limitations without requiring you to reproduce those attacks.

Investigative Claim to Prove

Before collecting evidence, your group must state one precise claim such as: *“We will demonstrate the differences in the authentication and key establishment process between WPA2-PSK and WPA3-SAE at the frame level, identifying which fields show the change in password-authenticated key exchange and what remains observable to a passive monitor in both cases.”* A WPA2-only comparison is equally valid if WPA3 infrastructure is unavailable — see the infrastructure note below.

Required Experiment

Capture and compare at least two relevant network modes, for example:

- WPA2-Personal.
- WPA3-SAE.
- WPA2/WPA3 transition mode.
- OWE / Enhanced Open, if supported.
- Open network as a baseline comparison only.

Scope Requirement

Do not attempt to analyze every security mode. Choose two modes and analyze them deeply. A narrow comparison with precise packet-level evidence is preferred over a broad comparison with superficial results.

Depending on your investigative claim, relevant evidence may include:

- Beacon RSN information.
- Association request and association response RSN elements.
- EAPOL 4-way handshake messages, when present.
- ANonce and SNonce.
- Replay counters.
- Key MIC, Key ACK, Install, Secure, and Encrypted Key Data flags.
- AKM suite selectors and cipher suite selectors.
- SAE commit/confirm frames, if analyzing WPA3-SAE.
- OWE Diffie-Hellman parameters, if analyzing OWE.

Required Security Analysis

Your report must answer:

- What security mode does the AP advertise?
- What security mode does the client select?
- Which fields show WPA2, WPA3, SAE, PSK, OWE, CCMP, GCMP, or PMF behavior?
- Where do ANonce and SNonce appear?
- What is the purpose of the replay counter?
- Which messages prove possession of key material?
- What would a replay, downgrade, or key-reinstallation attack attempt to manipulate?
- What can be observed passively, and what remains secret?

What to Expect

This is one of the most academically valuable tracks. Passive capture is sufficient and the analysis is reliable. Existing handshake-capture tools are common, but you must provide your own annotated interpretation and parser output.

Infrastructure Note — WPA3 and OWE

WPA3-SAE and OWE require an AP that explicitly supports those modes. Before selecting this track with the intention of analyzing WPA3 or OWE, verify that your AP supports the required mode and confirm this in writing with the instructor. A WPA2-only submission — comparing WPA2-Personal with an open baseline, or comparing with and without PMF — is fully sufficient for a strong submission and requires no special infrastructure.

5.3 Track 3: SSID/BSSID Identity and Evil Twin Trust Assumptions

Central Question

What identity does a Wi-Fi client actually authenticate: SSID, BSSID, AP, network profile, or key possession?

What This Track Investigates

This track connects Assignment 1 to a deeper protocol question. In an Evil Twin attack, the attacker imitates a known network. In this track, you investigate what information the client uses to decide “this is my network,” what is displayed to the user, and what is actually protected by the protocol.

Investigative Claim to Prove

Before collecting evidence, your group must state one precise claim such as: *“We will demonstrate that the SSID is visible in multiple unprotected frame types before any association takes place, that the BSSID is not cryptographically bound to the SSID in WPA2, and that a client will initiate association with a rogue AP presenting a known SSID without detecting the identity mismatch at the protocol level.”* Your report must return to this claim and prove or qualify it using frame-level evidence.

Required Experiment

Build a controlled lab using at least two AP configurations, for example:

- Same SSID, different BSSID.
- Different SSID, same credentials.
- Same SSID, different security mode.
- WPA2 network compared with WPA2/WPA3 transition mode.
- Legitimate-looking AP compared with a controlled rogue-looking AP.

Depending on your investigative claim, relevant evidence may include:

- Beacon frames.
- Probe requests and probe responses.
- Authentication frames.
- Association request and response.
- RSN negotiation.
- 4-way handshake, when present.
- Client reconnection or roaming behavior, if observed.

Required Security Analysis

Your report must answer:

- Where does the SSID appear in the trace?
- Where does the BSSID appear in the trace?
- What is cryptographically authenticated?
- Is the SSID itself authenticated in the way users intuitively assume?
- When does the client decide that a network matches a saved profile?
- What role do credentials play?
- What does Evil Twin exploit at the user/interface layer?
- What does SSID confusion exploit at the protocol-binding layer?

What to Expect

This is a strong continuation from Assignment 1. Any two WPA2-capable APs are sufficient — no special infrastructure is required. Public research code exists for SSID-confusion experiments and Evil Twin tools are common, but the deliverable here is not a copied tool. It is a controlled identity experiment and a trace-based explanation.

5.4 Track 4: MAC/PHY Metadata Leakage and Optional BFI

Central Question

What security-relevant information leaks below encryption?

What This Track Investigates

This track focuses on information that remains observable even when payload data is encrypted. It connects the MAC layer, PLCP/PMD concepts, monitor-mode metadata, radiotap headers, and optional beamforming feedback information (BFI).

Investigative Claim to Prove

Before collecting evidence, your group must state one precise claim such as: “*We will demonstrate that a passive observer with no decryption capability can distinguish at least two different physical or behavioral conditions using only metadata available in radiotap headers and 802.11 MAC fields, and we will identify which specific fields provide the distinguishing signal.*” Your report must return to this claim and prove or qualify it using captured evidence and script output.

Required Experiment

Compare at least two controlled physical or MAC-layer conditions, for example:

- Static client vs. moving client.
- Near client vs. far client.
- Different channels or bandwidths.
- Different AP placement.
- Idle client vs. active traffic generation.
- Empty room vs. benign movement, if using BFI.

Capture and analyze at least four of the following. **At least one of your four chosen items must come from the behavioral group** (marked with †) rather than from static radio parameters alone:

- RSSI / received signal strength.
- Channel and center frequency.
- Data rate or MCS index.
- Channel width.
- † Retry behavior and retry flag distribution.
- † Frame timing and inter-frame spacing.
- † Frame length distribution.
- † Traffic volume over time.
- Radiotap metadata fields.
- Beamforming feedback information (BFI), if available and ethically collected.

Required Security Analysis

Your report must answer:

- Which information remains visible despite WPA2/WPA3 encryption?
- Which fields come from 802.11 headers, and which come from radiotap or monitor metadata?
- What physical or behavioral condition can your evidence distinguish?
- What privacy or security conclusion is justified by your evidence?
- What conclusion is **not** justified?
- If using BFI: what is BFI, why does it exist, who sends it, and why can it leak information?

Hardware prerequisites for BFI Work

BFI capture requires a beamforming-capable 802.11ac or 802.11ax AP with transmit beamforming **enabled** in its settings, and a client that actually responds with compressed beamforming feedback. Verify both before committing to BFI as your evidence channel — a quick check with Wireshark or TShark display filters for VHT/HE compressed beamforming feedback or beamforming-report-related action frames can help confirm whether such frames are present. The exact field names may depend on the Wireshark version.

What to Expect

Basic radiotap and RSSI analysis is suitable for any group and requires only the course adapter in monitor mode — no special infrastructure. BFI analysis is advanced and hardware-dependent. Existing BFI extraction code exists, but a strong submission must still explain the frames, the extracted values, the experiment design, and the limits of inference.

6 Common Required Deliverables

Every group must submit the following components, regardless of track.

6.1 1. Research Report

Submit a PDF report with the following structure. The report must be **12 to 20 pages** in length, excluding figures, packet tables, and appendices.

1. **Title page:** Group members, IDs, selected track, date, hardware, OS, adapter, AP/router, client device.
2. **Investigative claim:** One precise paragraph stating what you set out to prove before collecting evidence.
3. **Evidence plan:** A short table explaining which packets, fields, parser outputs, tables, or plots are required to evaluate the investigative claim.

4. **Background:** Protocol concepts required to understand the investigation.
5. **Threat model:** Attacker position, capabilities, assumptions, and limitations.
6. **Lab setup:** Diagram and description of AP, client, monitor adapter, channels, bands, security modes, and software versions.
7. **Experiment design:** Baseline condition, modified condition, expected observation, and measurement method.
8. **Packet evidence:** Annotated packet table and screenshots or excerpts from Wireshark/TShark where useful.
9. **Parser / analysis output:** Explanation of your script and its output files.
10. **State-machine or sequence diagram:** Show the relevant protocol flow.
11. **Security analysis:** Explain what the evidence proves, what it does not prove, and why it matters.
12. **Limitations:** Explain what was not proven, what failed, and what hardware or driver limitations affected the work.
13. **Ethics and scope:** State how you ensured authorization and privacy protection.
14. **References:** Standards, documentation, research papers, and tools used.

6.2 2. Packet Captures

Submit the original raw capture and, if useful, a smaller filtered capture used for report annotation. Do not submit only screenshots. The examiner must be able to open the PCAP/PCAPNG and verify the packet numbers cited in the report.

- Format: .pcap or .pcapng.
- Minimum: at least two captures — baseline and modified condition.
- Name captures clearly, for example:

```
pmf_disabled_baseline.pcapng
pmf_required_modified.pcapng
wpa2_handshake_baseline.pcapng
wpa3_sae_modified.pcapng
ssid_same_name_different_bssid.pcapng
rssi_static_vs_movement.pcapng
```

6.3 3. Annotated Packet Table

Your report must contain a table similar to the following:

Frame	Time	Type/Subtype	Transmitter	Receiver	Important Fields	Interpretation
12	0.000	Beacon	AP	Broadcast	SSID, RSN IE, PMF flags	AP advertises security capabilities.
41	2.311	Association Request	STA	AP	RSN IE, selected AKM	Client requests association using selected security parameters.
52	2.490	EAPOL M1	AP	STA	ANonce, replay counter	AP starts the 4-way handshake.
53	2.492	EAPOL M2	STA	AP	SNonce, MIC	Client proves possession of derived key material.

Table 1: Example annotated packet table. Replace with your own evidence.

6.4 4. Parser or Analysis Script

Submit a small script that extracts the fields relevant to your track. The script does not need to be large, but it must be reproducible and understandable.

Acceptable choices include:

- Python with Scapy.
- Python with PyShark.
- TShark command pipeline with a short wrapper script.
- Other tools approved in advance.

The script should produce a structured output file such as:

- `frames.csv`
- `eapol_messages.json`
- `rsn_elements.csv`
- `radiotap_stats.csv`
- `bfi_features.csv`

Minimum Parser Requirement

Your parser must extract the fields needed to evaluate your investigative claim. At minimum, it must extract frame number or index, timestamp, transmitter, receiver, BSSID when relevant, frame type/subtype, and at least five claim-relevant fields that are used directly in your report.

6.5 5. Claim-to-Evidence Table

Every major technical claim must be tied to evidence. Include a table like this:

Claim	Evidence	Explanation
The AP advertises PMF capability.	Frame 12, RSN capabilities field	The RSN information element contains the PMF Capable bit.
The client selected WPA2-PSK with CCMP.	Frame 41, association request RSN IE	The AKM and pairwise cipher suite selectors indicate PSK and CCMP.
The observed EAPOL message is Message 1 of 4.	Frame 52, EAPOL key flags and nonce	The message contains an ANonce and no MIC from the client.

Table 2: Example claim-to-evidence table. Replace with your own claims and evidence.

6.6 6. Reproducibility Package

Submit a compressed archive or GitHub repository containing:

- Final report in PDF.
- All PCAP/PCAPNG files used in the report.
- Parser/analysis script.
- Parser output files.
- README file with setup and execution instructions.
- List of hardware and software versions.
- Any configuration files used for AP setup or capture setup, if applicable.

7 Allowed and Prohibited Tools

7.1 Allowed Tools

You may use standard analysis and system tools, including but not limited to:

- Wireshark, TShark, tcpdump
- Scapy, PyShark

- `iw`, `ip`, `iwconfig`, `rkill`
- `hostapd`, `dnsmasq`, `NetworkManager`, where relevant
- Python data-analysis libraries: `pandas`, `numpy`, `matplotlib`

7.2 Public Research Code

Public research code may be used only as a reference for understanding a paper, reproducing a parser concept, or designing an authorized experiment. If you use any public code, you must:

- Cite it clearly.
- Explain exactly what part you used.
- Explain what you wrote independently.
- Include your own evidence and your own analysis.

7.3 Prohibited Behavior

Tool and Conduct Restrictions

The following are prohibited unless explicitly approved:

- Running pre-built attack frameworks as the main submission.
- Submitting copied exploit code as your work.
- Capturing, storing, or analyzing traffic from uninvolved devices or networks.
- Attempting to recover real credentials, passwords, PINs, private messages, or personal data.
- Performing denial-of-service activity outside your controlled lab.
- Modifying captures, code, or report files after the submission deadline.

8 Required Hardware

A wireless network adapter capable of monitor-mode capture is required. The adapter distributed in the course is:

- **EDUP AX3000 / EP-AX1672**
- Chipset: MediaTek MT7921U family (confirmed via USB ID 0e8d:7961)
- Standard: Wi-Fi 6 (802.11ax), dual-band 2.4 GHz and 5 GHz
- Driver: `mt7921u`, mainline Linux kernel 5.18 and later

- Confirmed capabilities: monitor mode, radiotap headers, channel switching across 2.4 GHz and 5 GHz bands
- Do not rely on 6 GHz operation unless you have verified it on your own adapter, driver, regulatory domain, and AP. All four tracks are fully achievable on 2.4 GHz and 5 GHz.

Hardware Expectation

This assignment is designed so that passive monitor-mode capture and analysis are sufficient. Packet injection and successful exploit reproduction are not required.

If you use a virtual machine, verify that the USB adapter is passed through correctly and that monitor mode works before beginning the experiment. Run the following commands to confirm:

```
sudo ip link set <interface> down
sudo iw dev <interface> set type monitor
sudo ip link set <interface> up
sudo iw dev <interface> set channel 6 # 2.4 GHz, busy channel
sudo tcpdump -i <interface> -c 10 type mgt
```

You should see management frames (beacons, probe requests) immediately. If no frames appear within 15 seconds, also try channel 1 or 11 (2.4 GHz) or 36 (5 GHz). A silent monitor on every channel — not just one — indicates a real passthrough or driver problem.

9 Grading

Your assignment grade has two components:

1. **Base Grade:** Quality of the group submission.
2. **Defense Multiplier:** Individual frontal defense multiplier.

$$\text{Assignment 2 Final Grade} = \min(100, \text{A2 Base Grade} \times \text{A2 Defense Multiplier})$$

The Assignment 2 defense multiplier reflects each student's individual performance on the Assignment 2 portion of the combined defense session. The 1.3 ceiling allows a strong defense to recover some lost base points, but the final grade is capped at 100. Assignment 1 is graded under the same formula using its own separate multiplier; see the Assignment 1 document.

9.1 Base Grading Criteria

Category	Weight	Description
Experiment Design and Lab Control	15%	Clear research question, controlled setup, baseline and modified conditions, proper isolation, and ethical scope.
Quality of Captures and Evidence	20%	Relevant PCAPs, correct capture conditions, useful packet selection, and clean evidence connected to the claim.
Frame and Field Annotation	20%	Accurate explanation of 802.11 frame types, RSN/EAPOL/PMF/SAE/OWE/radiotap/BFI fields as relevant to the selected track.
Parser or Analysis Script	15%	Reproducible script that extracts meaningful fields and produces structured output used in the report.
Security Reasoning and Limitations	20%	Correct distinction between observation, inference, security claim, and limitation. Strong explanation of what the evidence proves and does not prove.
Report Clarity and Reproducibility	10%	Clear writing, organized submission, README, references, file naming, and ability to reproduce the analysis.

Table 3: Base grading criteria.

9.2 Frontal Defense

The frontal defense is conducted as a single Zoom session with the entire work group, lasting approximately 90–120 minutes per group. The session covers **both Assignment 1 and Assignment 2**, and each student is questioned individually within the group meeting. Prepare both submissions — captures, code, parser output, and report — and be ready to navigate either one when called on.

Each student receives **two separate defense multipliers** from this session: one applied to their Assignment 1 grade based on their answers about the Evil Twin tool, and one applied to their Assignment 2 grade based on their answers about the investigation, captures, parser, and analysis. A student may receive a strong multiplier on one assignment and a weak one on the other, reflecting where their individual performances were better.

During the defense you should expect to:

- Demonstrate and explain components of your Assignment 1 Evil Twin tool — wireless discovery, targeted disconnection, rogue AP operation, credential capture, and your defense component — and answer questions about why each piece works at the protocol level.
- Open your group’s capture file and navigate to specific frames named by the examiner.

- Explain what a named field contains, why it appears in that frame, and what its security significance is.
- Describe the protocol state at a specific point in the trace — what has been established, what has not yet been negotiated, and what an observer at that point would know.
- Justify one of the claims in your claim-to-evidence table using only your captures and your own words.
- Explain what your parser does, what a specific output line means, and why you chose to extract those fields.
- State what your evidence does **not** prove, and describe one concrete way you would improve the experiment if you repeated it.

The defense multiplier ranges from **0.0 to 1.3** and reflects each student's actual understanding and individual contribution to the work.

Category	Weight	What This Measures
Trace Comprehension	35%	Can you navigate your own capture, identify specific frames, and explain the important fields without consulting notes?
Protocol Knowledge	30%	Do you understand the relevant 802.11, RSN, PMF, WPA2/WPA3/OWE, MAC, or PHY concepts underlying your investigation?
Script and Analysis Understanding	20%	Can you explain your parser, its output files, and the reasoning behind your analysis decisions?
Limitations and Critical Thinking	15%	Can you state what your evidence does not prove, identify one weakness in your experiment design, and suggest a concrete improvement?

Table 4: Defense evaluation categories.

Defense Performance Can Make or Break Your Grade

The frontal defense is not a formality. A student who cannot navigate the group's captures, explain specific fields, or justify the claims in the submission may receive a multiplier of 0.5 or lower even if the group submission is strong. A student who demonstrates deep understanding beyond the submission may receive a multiplier above 1.0.

10 Examples of Good and Bad Claims

Weak Claim	Better Claim
“PMF is secure.”	“In our PMF-required capture, the RSN capabilities field indicates PMF Required, and the observed post-association deauthentication behavior differs from the PMF-disabled baseline. This supports the claim that robust management-frame protection is active in this configuration.”
“WPA3 is better than WPA2.”	“The WPA3-SAE capture contains SAE commit/confirm authentication frames before the 4-way handshake, while the WPA2-PSK capture proceeds from open authentication/association to EAPOL. This shows a different password-authenticated key-establishment process.”
“The Evil Twin worked.”	“The client selected a BSS based on SSID/profile behavior. Our trace shows where SSID and BSSID appear and which parts of the connection are later protected. This explains the trust assumption exploited by Evil Twin-style attacks.”
“Encryption hides everything.”	“Payload contents are encrypted, but radiotap metadata, timing, frame length, transmitter/receiver addresses in many frames, and management-frame metadata remain observable. Our script shows that condition A and condition B produce distinguishable RSSI/timing distributions.”

Table 5: Examples of weak and stronger evidence-based claims.

11 Submission Guidelines

- **Track selection deadline:** TBA — submit by email with subject line **CWN 2026 — Track Selection**. Confirmed by reply only. Groups may choose the same track, but the investigative claim, experiment design, and capture scenario must be distinct.
- **Submission deadline:** TBA (approximately 7 weeks from distribution).
- **Submission method:** Compressed archive or GitHub repository link, according to course instructions.
- **Required archive name:** Assignment2_GroupXX.zip or equivalent.

SEVERE WARNING — No Changes After Deadline

Do not make any changes to the repository or submitted files after the final deadline. Unauthorized post-deadline changes will disqualify the group’s submission.

11.1 Required Submission Components

1. **Report PDF** — complete report per required structure, including packet tables, diagrams, claim-to-evidence table, and references.

2. **PCAP/PCAPNG files** — baseline and modified-condition captures, plus any additional captures used in the report.
3. **Parser or analysis code** — script files with clear comments and no unrelated code.
4. **Output files** — CSV/JSON/text files generated by your script, plus plots if relevant.
5. **README** — group details, hardware and software versions, capture setup, script execution instructions, known limitations, and ethical scope statement.

12 Suggested Technical References

Your report must cite at least one standards-level source, one tool or dissector reference, and one research or security reference relevant to your selected track. Blog posts may be used only as supporting material, not as the primary technical reference.

12.1 General Protocol and Tool References

- IEEE 802.11 standard family and 802.11i/RSN material, available through university library IEEE Xplore access.
- Wireshark WLAN display-filter reference:
<https://www.wireshark.org/docs/dfref/w/wlan.html>
- Wireshark RSNA EAPOL display-filter reference:
https://www.wireshark.org/docs/dfref/w/wlan_rsna_eapol.html
- Scapy 802.11 / Dot11 layer documentation:
<https://scapy.readthedocs.io/en/latest/api/scapy.layers.dot11.html>
- RFC 8110 — Opportunistic Wireless Encryption:
<https://www.rfc-editor.org/rfc/rfc8110>

12.2 Research Sources for Deeper Analysis

- KRACK — Key Reinstallation Attacks: <https://www.krackattacks.com/>
- Dragonblood — WPA3 Dragonfly/SAE analysis: <https://wpa3.mathyvanhoef.com/>
- FragAttacks — Fragmentation and Aggregation Attacks: <https://www.fragattacks.com/>
- SSID Confusion paper: <https://papers.mathyvanhoef.com/wisec2024.pdf>
- BFI / Wi-BFI project (if choosing the BFI extension): <https://github.com/kfoysalhaque/Wi-BFI>

12.3 Hardware and Environment References

- EDUP EP-AX1672 product information:
<https://www.szedup.com/usb-adaptersBr3/EP1672.html>
- Linux USB Wi-Fi adapter chipset and driver reference:
<https://github.com/morrownr/USB-WiFi>
- DragonOS: <https://cemaxecuter.com/>
- DragonOS community documentation:
<https://github.com/DragonOS-Community>

13 Contact Information

If you have questions or concerns, contact the instructor. Please include **CWN 2026 — Assignment 2** in the email subject.

- **Instructor:** Dr. Eyal Berliner
- **Email:** eyalbe@ariel.ac.il
- **Office Hours:** By appointment

Remember: If there is doubt, there is no doubt that you should contact me—either by email, in class, or during office hours.

Best of luck,

Eyal